

Sysadmin Prep Lab Guide

A full step-by-step, hands-on roadmap to build real system administration skills before your internship starts

15 weeks • ~8–10 hrs/week • built for a 100GB disk budget

Covers: Windows Server & Active Directory, Linux administration, networking & firewalls, scripting/automation, monitoring & security, and CompTIA cert prep

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PHASE 0

Before You Start

Day 1 | Get your lab environment ready to build on

Step 0.1 — Install a hypervisor

Pick one. VirtualBox is simplest for a first lab; Proxmox is closer to what real datacenters run (install it as the PC's main OS if you're willing to dedicate the machine, or nest it in VirtualBox if not).

1. Go to [virtualbox.org](https://www.virtualbox.org) and download the installer for your OS.
2. Run the installer with default options (it will briefly interrupt your network connection during install — this is normal).
3. Also install the "VirtualBox Extension Pack" from the same download page — adds USB/network features you'll want later.

Step 0.2 — Download your install media (do this once, then delete ISOs after use)

- **Ubuntu Server LTS** — ubuntu.com/download/server — ~2GB
- **Windows Server 2022 Evaluation** — Microsoft Evaluation Center (free, 180-day, fully legal, no license key needed) — ~5GB
- **AlmaLinux Minimal** — almalinux.org/get-almalinux — ~2GB (used later for RHCSA-style practice)

Note: Save ISOs to a dedicated folder like C:\labisos so you remember to delete them once installed.

Step 0.3 — Create a lab network in VirtualBox

1. Open VirtualBox → File → Tools → Network Manager.
2. Create a new "NAT Network" (gives your VMs internet access AND lets them talk to each other) — name it LabNet, subnet 10.10.10.0/24.
3. You'll attach every lab VM's network adapter to this LabNet instead of the default NAT — this is what makes your VMs able to see each other.

Step 0.4 — Set up your accounts

1. Create a free GitHub account if you don't have one.
2. Create a new repository called something like sysadmin-lab-journal — this is where every phase's write-up will go.
3. Create a free Oracle Cloud account (Free Tier) and a free Azure account — you won't need them immediately, but Phase 6 may push your 100GB, and it's easier to have these ready.

CHECKPOINT — VirtualBox installed, 3 ISOs downloaded, LabNet created, GitHub repo exists.

PHASE 1

Foundations Refresh

Weeks 1–2 | Rebuild the fundamentals so everything after this sticks

Step 1.1 — Subnetting, daily, until it's automatic

Do 15 minutes a day on a free subnetting practice site or app. You should be able to answer "what's the network address, broadcast address, and usable host range for 192.168.14.37/27" in under 30 seconds without a calculator by the end of week 2.

Step 1.2 — Review the concepts you'll be building this week

- OSI model & TCP/IP model — explain each layer out loud, in your own words
- What DNS actually resolves and why it's a hierarchy, not a flat list
- What DHCP negotiates (the DORA process: Discover, Offer, Request, Acknowledge)
- What Active Directory is solving (centralized identity & policy, not just "a list of users")

Step 1.3 — Build your first VM: Ubuntu Server

1. VirtualBox → New. Name it **lab-linux-01**. Type: Linux, Version: Ubuntu (64-bit).
2. Memory: 2048 MB. Processors: 2.
3. Hard disk: create a new VDI, **dynamically allocated**, 25GB (this is a ceiling, not real usage — a fresh install only consumes ~3–4GB).
4. Before starting: Settings → Network → Adapter 1 → attach to **NAT Network** → LabNet.
5. Settings → Storage → click the empty disk icon → choose your Ubuntu Server ISO.
6. Start the VM. Walk through the installer: language → keyboard → network (leave DHCP default) → skip proxy → use default storage layout → set your profile (hostname: **lab-linux-01**) → on the "Featured Server Snaps" screen, check **OpenSSH server** — this is the one screen people forget and then can't SSH in later.
7. Let it finish, reboot, remove the ISO from Settings → Storage once it boots to a login prompt.

Step 1.4 — Set a static IP and confirm SSH access

lab-linux-01 (run inside the VM console)

```
# Inside the VM, find your current DHCP-assigned IP:
ip a

# Edit netplan config (filename may differ slightly):
sudo nano /etc/netplan/50-cloud-init.yaml
```

/etc/netplan/50-cloud-init.yaml

```
network:
  ethernets:
    enp0s3:
      dhcp4: no
      addresses: [10.10.10.11/24]
      routes:
        - to: default
          via: 10.10.10.1
      nameservers:
        addresses: [8.8.8.8]
  version: 2
```

lab-linux-01

```
sudo netplan apply
ip a    # confirm 10.10.10.11 is now assigned
```

Now from your host PC's terminal (or PuTTY on Windows):

host machine

```
ssh yourusername@10.10.10.11
```

Note: If SSH times out, double-check the VM's network adapter is attached to LabNet, not the default NAT adapter.

CHECKPOINT — One Ubuntu Server VM running with a static IP, reachable via SSH from your host machine.

PHASE 2

Windows Server & Active Directory

Weeks 3–4 | Stand up the backbone of most corporate networks

Step 2.1 — Build the Windows Server VM

1. VirtualBox → New. Name: **lab-dc-01**. Type: Windows, Version: Windows 2022 (64-bit).
2. Memory: 4096 MB minimum. Processors: 2.
3. Disk: dynamically allocated, 40GB.
4. Network adapter: attach to **NAT Network** → LabNet (same as before).
5. Mount the Windows Server Evaluation ISO and start the VM.
6. In the installer, choose **Windows Server 2022 Standard Evaluation (Server Core)** — NOT the "Desktop Experience" option. Server Core has no GUI — this is intentional and matches real environments.
7. Accept license, choose Custom install, select the virtual disk, let it install and reboot.
8. On first boot you'll be dropped into a black screen prompting a password change — set a strong local Administrator password and remember it.

Step 2.2 — Configure it using sconfig (Server Core's menu-driven config tool)

You'll land on a command prompt. Type:

lab-dc-01

```
sconfig
```

1. Option 2 (Computer Name) → rename to **DC01** → reboot when prompted.
2. After reboot, run sconfig again → Option 8 (Network Settings) → select your adapter → Option 1 (Set Network Adapter Address) → Static → IP 10.10.10.10, subnet 255.255.255.0, gateway 10.10.10.1.
3. Still in Network Settings → Option 2 (Set DNS Servers) → primary 127.0.0.1 (it will become its own DNS server shortly).
4. Exit sconfig (option 15) back to command prompt.

Step 2.3 — Install Active Directory Domain Services (PowerShell)

Server Core drops you to cmd by default — type `power shell` and hit Enter first.

lab-dc-01 (PowerShell)

```
Install-WindowsFeature AD-Domain-Services -IncludeManagementTools

Install-ADDSForest `
  -DomainName "lab.local" `
  -DomainNetbiosName "LAB" `
  -InstallDns:$true `
  -SafeModeAdministratorPassword (ConvertTo-SecureString "YourSafeModePass1!" -AsPlainText -Force) `
  -Force:$true
```

The server reboots automatically once this completes. Log back in as LAB\Administrator.

Step 2.4 — Install DHCP and scope out your lab subnet

lab-dc-01 (PowerShell)

```
Install-WindowsFeature DHCP -IncludeManagementTools

Add-DhcpServerv4Scope -Name "LabScope" -StartRange 10.10.10.100 `
  -EndRange 10.10.10.200 -SubnetMask 255.255.255.0 -State Active

Set-DhcpServerv4OptionValue -ScopeId 10.10.10.0 -Router 10.10.10.1 `
  -DnsServer 10.10.10.10

netsh dhcp add securitygroups
Restart-Service dhcpserver
```

Step 2.5 — Build your OU structure, users, and groups

lab-dc-01 (PowerShell)

```
New-ADOrganizationalUnit -Name "Sales" -Path "DC=lab,DC=local"
New-ADOrganizationalUnit -Name "IT" -Path "DC=lab,DC=local"
New-ADOrganizationalUnit -Name "Finance" -Path "DC=lab,DC=local"

New-ADUser -Name "Jane Doe" -SamAccountName "jdoe" `
  -Path "OU=Sales,DC=lab,DC=local" `
  -AccountPassword (ConvertTo-SecureString "Passw0rd!23" -AsPlainText -Force) `
  -Enabled $true

New-ADGroup -Name "Sales-Team" -GroupScope Global -Path "OU=Sales,DC=lab,DC=local"
Add-ADGroupMember -Identity "Sales-Team" -Members "jdoe"
```

Repeat New-ADUser for 4–9 more test accounts across your OUs — this is the kind of repetitive task worth wrapping in a small script (you'll do exactly that in Phase 5).

Step 2.6 — Create and link a Group Policy Object

lab-dc-01 (PowerShell)

```
Install-WindowsFeature GPMC
New-GPO -Name "Sales-Password-Policy" | New-GPLink -Target "OU=Sales,DC=lab,DC=local"
```

For editing GPO settings visually, install RSAT's Group Policy Management Console on a client VM or a Windows host machine and connect to DC01 remotely — editing GPOs blind via PowerShell registry paths is possible but

painful as a first exercise.

Step 2.7 — Join a client to the domain

Build a lightweight Windows 10/11 VM (or reuse a spare machine), attach it to LabNet, set its DNS server to 10.10.10.10, then:

client VM (PowerShell, run as local admin)

```
Add-Computer -DomainName "lab.local" -Credential LAB\Administrator -Restart
```

Step 2.8 — Run the full ticket simulation

Now do this end-to-end, timing yourself: "New employee Jane Doe joins Sales. Create her account, add her to Sales-Team, confirm the GPO applies when she logs in from the client VM, and map a shared drive for her." Write down every step you took, including anything that went wrong.

CHECKPOINT — Working domain with DNS + DHCP, 3 OUs, 5+ users, 1 linked GPO, a domain-joined client. Documented on GitHub.

PHASE 3

Linux Administration

Weeks 5–6 | The majority of servers in the world run Linux

Step 3.1 — Users, groups, and permissions

lab-linux-01

```
sudo useradd -m -s /bin/bash bob
sudo passwd bob
sudo groupadd devteam
sudo usermod -aG devteam bob

# Permissions in octal notation:
chmod 750 /some/directory      # rwxr-x---
chown bob:devteam /some/directory
```

Step 3.2 — sudo access, the safe way

lab-linux-01

```
sudo visudo
# add a line granting a group sudo rights instead of editing per-user:
%devteam ALL=(ALL) ALL
```

Step 3.3 — Package management

lab-linux-01

```
sudo apt update && sudo apt upgrade -y
sudo apt install nginx samba -y

# On your AlmaLinux VM later, the equivalent is:
sudo dnf update -y
sudo dnf install nginx samba -y
```

Step 3.4 — systemd: the service manager you'll live in

lab-linux-01

```
sudo systemctl status nginx
sudo systemctl stop nginx
sudo systemctl start nginx
sudo systemctl enable nginx      # start automatically on boot
sudo journalctl -u nginx -f      # live-tail its logs
```

Step 3.5 — Cron: schedule a real task

lab-linux-01

```
crontab -e
# add this line to log disk usage every night at 1am:
0 1 * * * df -h >> /home/bob/disk-usage.log
```

Step 3.6 — SSH hardening

lab-linux-01

```
ssh-keygen -t ed25519 -C "lab-key"           # run on your HOST machine
ssh-copy-id bob@10.10.10.11                 # copies your public key to the VM

sudo nano /etc/ssh/sshd_config
# PermitRootLogin no
# PasswordAuthentication no
# Port 2222

sudo systemctl restart sshd
```

Note: Test the new key-based login in a SEPARATE terminal window before closing your current session — if something's misconfigured, you want a working session still open to fix it.

Step 3.7 — Stand up a web server and a file share

lab-linux-01

```
sudo apt install nginx -y
sudo systemctl enable --now nginx
# visit http://10.10.10.11 from your host browser to confirm

sudo apt install samba -y
sudo nano /etc/samba/smb.conf
```

/etc/samba/smb.conf (append at the bottom)

```
[labshare]
path = /srv/labshare
browseable = yes
writable = yes
guest ok = no
```

lab-linux-01

```
sudo mkdir -p /srv/labshare
sudo smbpasswd -a bob
sudo systemctl restart smbd
# from your Windows client VM, browse to: \\10.10.10.11\labshare
```

Step 3.8 — Break it on purpose, then fix it

- Stop nginx mid-session, then diagnose using only systemctl status and journalctl — write down your diagnostic steps

- Fill the disk on a throwaway test VM with `dd if=/dev/zero of=/tmp/fill bs=1M count=2000`, watch what breaks, then recover space
- Misconfigure a permission (e.g. `chmod 000` a file nginx needs) and diagnose the resulting error from logs alone

Note: *These self-inflicted incidents are exactly the stories you'll tell in interviews. Write each one up: what broke, how you found out, what you changed.*

CHECKPOINT — Linux server reachable on the lab network, serving a page, sharing a folder, with 3 documented self-caused incidents and fixes.

PHASE 4

Networking Deep Dive

Weeks 7–8 | Understand the network, don't just plug into it

Step 4.1 — Build a pfSense router/firewall VM

1. Download the pfSense CE ISO from pfsense.org.
2. VirtualBox → New → name **lab-fw-01**, type BSD, 1024MB RAM, 8GB dynamic disk.
3. Give it **two** network adapters: Adapter 1 = NAT (this is your "WAN", internet-facing), Adapter 2 = Internal Network named **LabLAN** (this becomes your protected lab segment).
4. Boot the ISO, accept defaults through the installer, let it reboot, remove the ISO.

Step 4.2 — Initial pfSense configuration

On boot you'll get a console menu assigning WAN/LAN interfaces automatically. Confirm WAN = Adapter 1, LAN = Adapter 2. LAN defaults to 192.168.1.1/24 — from the console menu, option 2, change it to match your lab: 10.10.10.1/24, DHCP range 10.10.10.100–10.10.10.199.

***Note:** From here on, re-attach your existing lab VMs' network adapters to LabLAN instead of the old NAT Network — pfSense now becomes their gateway.*

Access the web GUI from your host machine (attach your host's networking so it can reach 10.10.10.1, or add a host-only adapter) at <https://10.10.10.1> and complete the setup wizard: hostname, DNS (8.8.8.8), timezone, set the admin password.

Step 4.3 — Create VLANs to segment servers from clients

1. pfSense GUI → Interfaces → VLANs → Add.
2. Create VLAN 10, description "Servers", parent interface = LAN.
3. Create VLAN 20, description "Clients", parent interface = LAN.
4. Interfaces → Assignments → assign both new VLAN interfaces, enable them, give each a subnet (e.g. 10.10.20.1/24 for Servers, 10.10.30.1/24 for Clients).
5. Move your DC and Linux server's virtual NICs onto the Servers VLAN; move client VMs onto the Clients VLAN.

Step 4.4 — Write firewall rules between the segments

Firewall → Rules → Clients tab. Add a rule: block Clients VLAN from reaching the Servers VLAN entirely, then add a second, more specific rule ABOVE it permitting Clients to reach the DC only on ports 53 (DNS), 88 (Kerberos), 389 (LDAP). Rule order matters in pfSense — first match wins.

***Note:** This single exercise — 'clients can talk to the domain controller but nothing else on the server segment' — is a genuinely realistic enterprise firewall pattern.*

Step 4.5 — Start Network+ study in parallel

- Use a free structured video course (e.g. Professor Messer's Network+ series) covering the current objectives
- Budget 30–45 min/day, 5 days/week
- After each domain, do 15–20 practice questions on that domain specifically before moving on

CHECKPOINT — Segmented lab network with VLANs, documented firewall rules, and steady Network+ practice-exam scores logged weekly.

PHASE 5

Automation & Scripting

Weeks 9–10 | This is what separates an admin from a button-clicker

Step 5.1 — PowerShell: bulk-create AD users from a CSV

Create-BulkUsers.ps1

```
# users.csv:
# Name,SamAccountName,OU,Password
# John Smith,jsmith,IT,Passw0rd!23
# Amy Tan,atan,Finance,Passw0rd!23

Import-Csv .\users.csv | ForEach-Object {
    New-ADUser -Name $_.Name `
        -SamAccountName $_.SamAccountName `
        -Path "OU=$(($_.OU),DC=lab,DC=local" `
        -AccountPassword (ConvertTo-SecureString $_.Password -AsPlainText -Force) `
        -Enabled $true
    Write-Host "Created $(($_.Name))"
}
```

Step 5.2 — PowerShell: report inactive accounts

Find-InactiveAccounts.ps1

```
$cutoff = (Get-Date).AddDays(-30)
Get-ADUser -Filter {LastLogonTimestamp -lt $cutoff -and Enabled -eq $true} `
    -Properties LastLogonTimestamp |
    Select-Object Name, SamAccountName,
        @{N='LastLogon';E={[DateTime]::FromFileTime($_.LastLogonTimestamp)}} |
    Export-Csv .\inactive-accounts.csv -NoTypeInformation
```

Step 5.3 — Bash: nightly backup script

backup.sh

```
#!/bin/bash
SRC="/srv/labshare"
DEST="/backups"
STAMP=$(date +%Y%m%d-%H%M)
mkdir -p "$DEST"
tar -czf "$DEST/labshare-$STAMP.tar.gz" "$SRC"
find "$DEST" -mtime +7 -name '*.tar.gz' -delete # keep 7 days
echo "Backup complete: labshare-$STAMP.tar.gz"
```

lab-linux-01

```
chmod +x backup.sh
crontab -e
# 0 2 * * * /home/bob/backup.sh >> /home/bob/backup.log 2>&1
```

Step 5.4 — Bash: log rotation sanity check

check-log-size.sh

```
#!/bin/bash
LOG="/var/log/nginx/access.log"
SIZE=$(stat -c%s "$LOG")
LIMIT=$((50*1024*1024)) # 50MB
if [ "$SIZE" -gt "$LIMIT" ]; then
    mv "$LOG" "$LOG.$(date +%Y%m%d)"
    systemctl reload nginx
    echo "Rotated: log exceeded 50MB"
fi
```

Step 5.5 — Optional stretch: Ansible

Ansible control node

```
sudo apt install ansible -y

# inventory.ini
[labservers]
10.10.10.11
10.10.10.12

# playbook.yml
- hosts: labservers
  become: yes
  tasks:
    - name: ensure nginx is installed
      apt:
        name: nginx
        state: present
```

run it

```
ansible-playbook -i inventory.ini playbook.yml
```

CHECKPOINT — 4-5 scripts pushed to GitHub, each with a short README explaining the real problem it solves.

PHASE 6

Security & Monitoring

Weeks 11–12 | Harden what you built, then watch it

Step 6.1 — Hardening pass across your lab

- Disable any service you're not actually using on each VM (systemctl list-unit-files --state=enabled to review)
- Apply all pending OS updates across every VM
- Enable a host firewall on the Linux boxes: `sudo ufw allow OpenSSH` && `sudo ufw allow 'Nginx Full'` && `sudo ufw enable`
- Review your pfSense rules from Phase 4 — remove anything overly permissive

Step 6.2 — Install Docker (keeps monitoring lightweight, space-wise)

lab-linux-01, or a new small VM: lab-mon-01

```
curl -fsSL https://get.docker.com | sudo sh
sudo usermod -aG docker $USER
# log out/in for the group change to apply
```

Step 6.3 — Deploy Zabbix via Docker Compose

docker-compose.yml

```

version: '3'
services:
  zabbix-db:
    image: mysql:8.0
    environment:
      MYSQL_DATABASE: zabbix
      MYSQL_USER: zabbix
      MYSQL_PASSWORD: zabbixpass
      MYSQL_ROOT_PASSWORD: rootpass
  zabbix-server:
    image: zabbix/zabbix-server-mysql:latest
    environment:
      DB_SERVER_HOST: zabbix-db
      MYSQL_DATABASE: zabbix
      MYSQL_USER: zabbix
      MYSQL_PASSWORD: zabbixpass
    ports: ["10051:10051"]
  zabbix-web:
    image: zabbix/zabbix-web-nginx-mysql:latest
    environment:
      DB_SERVER_HOST: zabbix-db
      MYSQL_DATABASE: zabbix
      MYSQL_USER: zabbix
      MYSQL_PASSWORD: zabbixpass
      ZBX_SERVER_HOST: zabbix-server
    ports: ["8080:8080"]

```

lab-mon-01

```

docker compose up -d
# visit http://<lab-mon-01 ip>:8080 (default login Admin / zabbix)

```

Step 6.4 — Add your lab VMs as monitored hosts

on each Linux VM you want monitored

```

sudo apt install zabbix-agent -y
sudo nano /etc/zabbix/zabbix_agentd.conf
# Server=<lab-mon-01 ip>
# Hostname=lab-linux-01
sudo systemctl restart zabbix-agent

```

Then in the Zabbix web UI: Data collection → Hosts → Create host → match the hostname, assign the "Linux by Zabbix agent" template, and confirm data starts flowing in within a minute or two.

Step 6.5 — Shift study focus to Security+

- This is your priority cert — it's the one requested most often in entry-level job postings
- Break the objectives into the 5 official domains and allocate specific days to each
- 45–60 min/day, and take a full-length practice exam at the end of week 12
- Log every practice score in a simple spreadsheet so you can see the trend

CHECKPOINT — Monitoring dashboard tracking your lab's VMs, hardening pass complete, Security+ practice scores trending toward 85%+.

PHASE 7

Certification Push

Weeks 13–14 | Convert practice into a booked (or passed) exam

Step 7.1 — Target your weak domains

Go back through every practice exam you've logged. Sort by domain, not just overall score — you want to know specifically which 1–2 domains are dragging you down, and spend disproportionate time there.

Step 7.2 — Full-length practice exams, timed

1. Take a full practice exam under real time pressure (90 questions, ~90 minutes) — no pausing, no notes.
2. Review every wrong answer and explain to yourself, out loud, why the correct answer is correct — not just what it is.
3. Repeat 2–3 times across the two weeks, aiming for consistent 85%+ before booking.

Step 7.3 — Book the exam

1. Register through Pearson VUE (CompTIA's testing partner) — choose an in-person test center or online proctored, whichever you're more comfortable with.
2. Bring a valid government ID matching your registration name exactly.
3. For online proctoring: test your webcam/mic beforehand and clear your desk completely — proctors are strict about a clean workspace.

Note: *If your practice scores aren't consistently above 85% by the end of week 14, it's genuinely fine to push the exam date back a couple weeks rather than sit it underprepared — a passed cert a bit later beats a failed attempt on your record.*

CHECKPOINT — Security+ passed, or booked with strong, consistent practice scores documented.

PHASE 8

Portfolio & Internship-Ready Polish

Week 15 | Package everything so it's easy for someone else to see

Step 8.1 — Organize your GitHub repo

Suggested repo structure

```
sysadmin-lab-journal/  
  README.md          <- the story: what you built and why, in plain language  
  ad-lab/             <- Phase 2 write-up + screenshots  
  linux-lab/          <- Phase 3 write-up + screenshots  
  network-lab/        <- Phase 4 write-up + firewall rule screenshots  
  scripts/            <- every script from Phase 5, each with its own README  
  monitoring/         <- Phase 6 dashboard screenshots + hardening notes
```

Step 8.2 — Resume bullet points, drawn straight from your lab

- Deployed and administered a multi-VM Windows/Linux lab environment including Active Directory, DNS, and DHCP
- Wrote PowerShell and Bash automation scripts to handle bulk user provisioning, inactive-account reporting, and backups
- Configured network segmentation and firewall policy using VLANs and pfSense
- Deployed a Zabbix monitoring stack via Docker to track infrastructure health
- CompTIA Security+ (passed / in progress, expected [month])

Step 8.3 — Prepare STAR-format stories from your incidents

Pick your best 3–4 "broke it, fixed it" moments from Phases 3, 4, and 6. For each, write out: **Situation** (what were you doing), **Task** (what needed to work), **Action** (exactly what you checked/changed, in order), **Result** (what you learned). This is precisely the format most interviewers are listening for, even when they don't say so.

Step 8.4 — Mock interview

1. Record yourself (phone camera is fine) explaining your lab in under 3 minutes, as if to a hiring manager who's never seen it.
2. Watch it back once — you'll immediately spot filler words and places you rushed.
3. Ask a friend or senior to run through 5–6 common sysadmin interview questions with you out loud, not just in your head.

CHECKPOINT — GitHub polished, resume updated, 3-4 STAR stories ready, one mock interview done. Ready for day one.

Appendix

Reference | Weekly time guide and space-saving cheat sheet

Weekly time guide

Weeks	Focus	Hrs/wk
1-2	Fundamentals	8
3-4	Windows Server / Active Directory	10
5-6	Linux administration	10
7-8	Networking + Network+ study	10
9-10	Scripting & automation	8
11-12	Security, monitoring + Security+ study	10
13-14	Certification push	10
15	Portfolio & interview prep	6

Space-saving cheat sheet (100GB budget)

- Always choose dynamically allocated virtual disks, never fixed-size
- Install Server Core instead of Desktop Experience wherever the step doesn't require a GUI
- Delete install ISOs once a VM is built and working
- Delete old VM snapshots once you're past that checkpoint — don't let them stack up
- If space gets tight from Phase 6 onward, shift that workload to your free Oracle Cloud or Azure account instead of local disk

Rough disk footprint per VM (dynamically allocated)

VM	Typical real usage
Ubuntu Server (lab-linux-01)	3-6 GB
AlmaLinux minimal	5-8 GB
Windows Server Core (lab-dc-01)	10-15 GB
pfSense (lab-fw-01)	2-4 GB
Monitoring VM (lab-mon-01, via Docker)	5-10 GB

Even running all five lab VMs at once, realistic total usage lands around 30–45GB — comfortably inside your 100GB with room to spare.